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Extended Bergman Minimal Model Implementation in Modelica

Glucose - Insulin dynamics for Diverse Patient Profiles

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Goal: Modeling glucose-insulin regulation in a physiologically interpretable dynamic framework to simulate healthy metabolism, different stages of diabetes, and the effects of insulin therapy and meals using Modelica.

Base Model: Bergman Minimal Model

$$\dot{G} = \frac{1}{60}\left(-p_1(G - G_b) - XG + \frac{D}{V_g}\right) \rightarrow \textbf{G}\text{: blood glucose concentration}$$

$$\dot{X} = \frac{1}{60}\left(-p_2 X + p_3(I - I_b)\right) \rightarrow \textbf{X}\text{: delayed insulin action in peripheral tissues}$$

$$\dot{I} = \frac{1}{60}\left(-k_I I + \frac{\text{insulin flow}}{V_i}\right) \rightarrow \textbf{I}\text{: plasma insulin concentration}$$

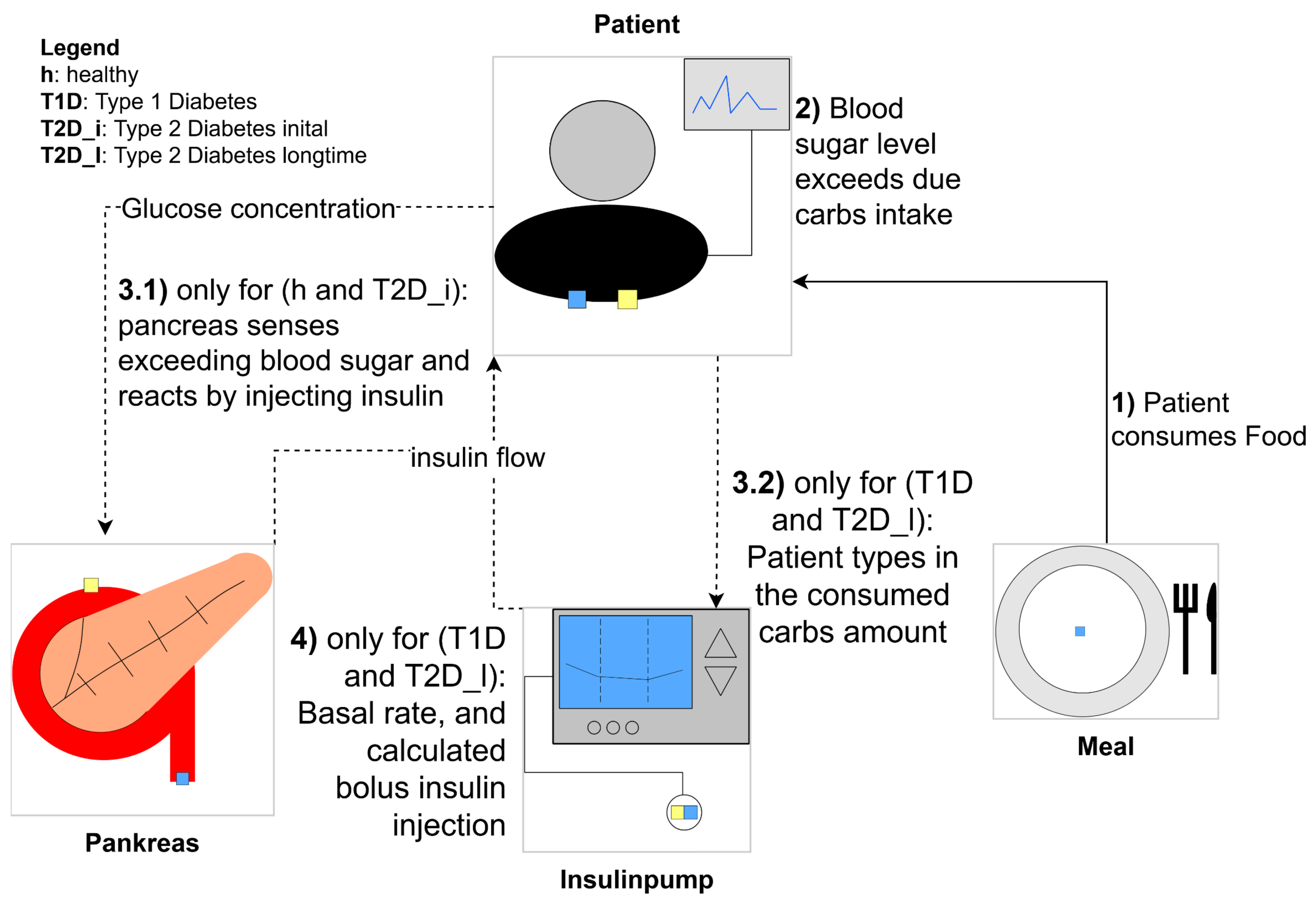


Figure 1: simulated process

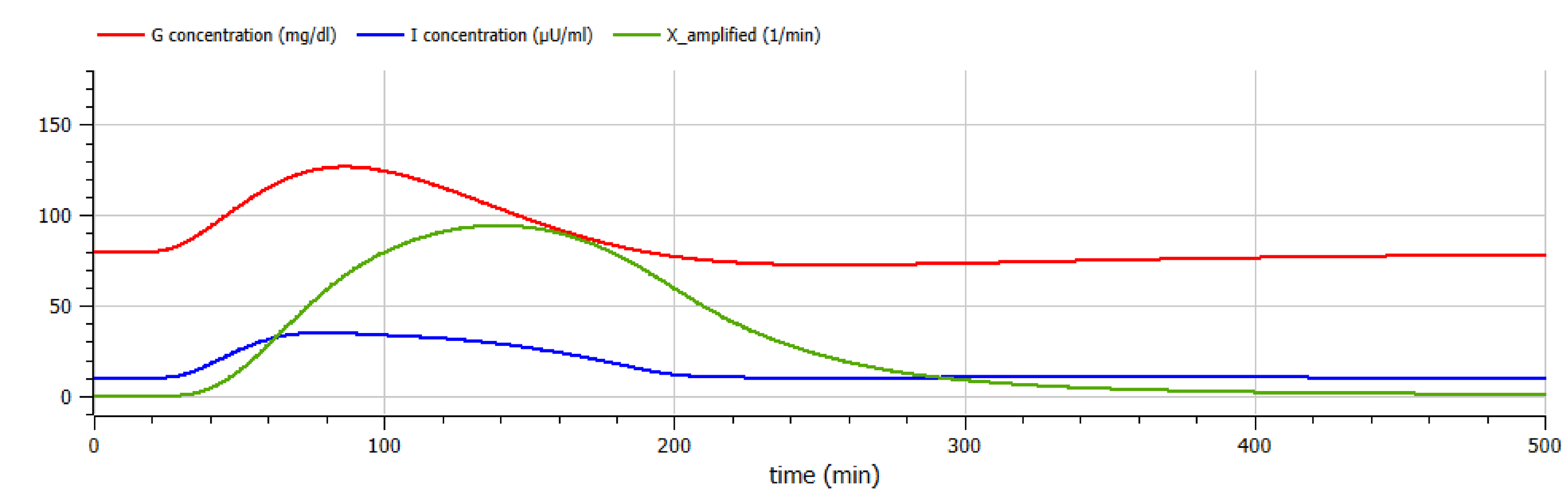
Model Extension: Explicit Insulin Secretion and Therapy

In the standard Bergman minimal model, insulin dynamics are not explicitly represented, while our extended model accounts for insulin delivery via an insulin flow.

- **Healthy pancreas:** biphasic endogenous insulin secretion
- **Type 1 diabetes:** no endogenous secretion → external insulin required
- **Type 2 diabetes (initial):** compensatory hyperinsulinemia with reduced insulin sensitivity; endogenous secretion is initially sufficient for glucose control
- **Type 2 diabetes (long-term):** reduced secretion capacity and slower dynamics → external insulin therapy often required

Model and Simulation Results

Healthy Patient



Type 2 Diabetes longtime

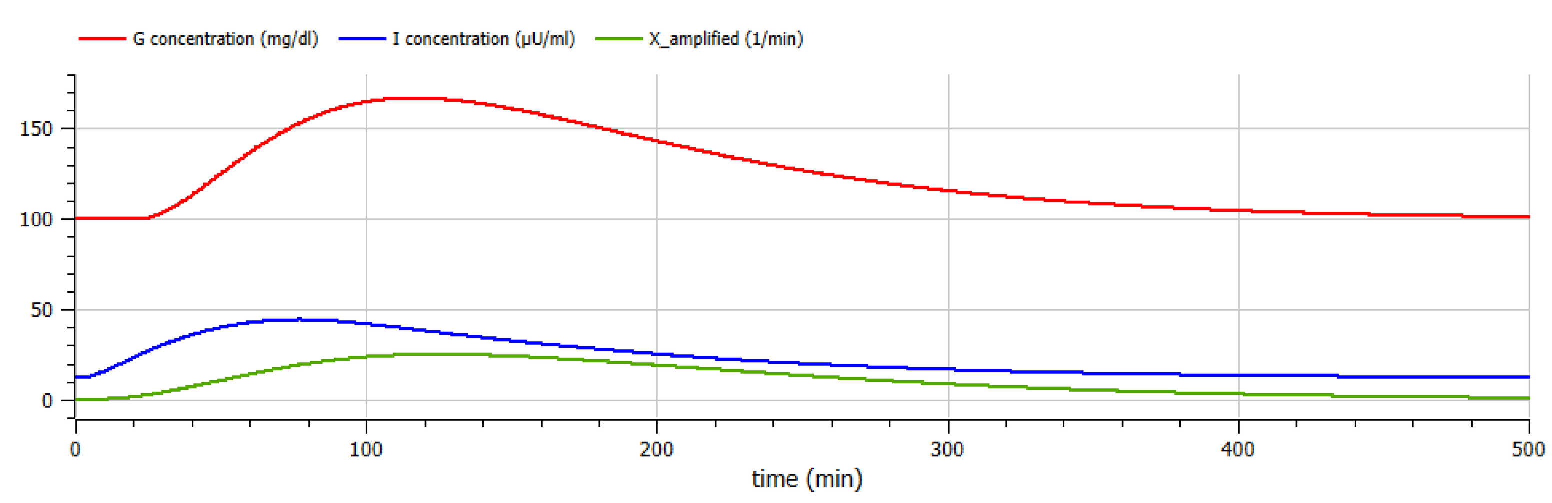


Figure 2: Glucose, Insulin concentration and X (insulin action) for different patient profiles when consuming the same meal

Outlook:

- Development of a hybrid closed-loop solution with automatic basal rate error correction while maintaining manual meal (bolus) input.
- The Bergman-based formulation assumes linear glucose dynamics around a fixed basal state. Future extensions may include nonlinear saturation effects and additional hormonal regulation to improve physiological realism

References:

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[3] Hall, J. E. (2021). Insulin, glucagon, and diabetes mellitus. *Guyton and Hall textbook of medical physiology* (14th ed., pp. 973–989). Elsevier.

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